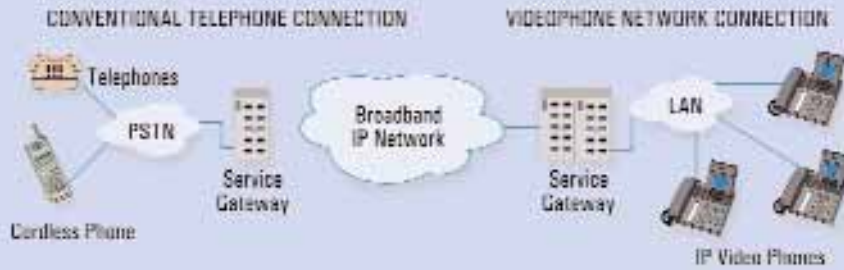


## Convergence at Home



Most homes today have regular PSTN telephone connections. As telecom service providers change over to data networks and broadband connectivity makes inroads, we will see a considerable amount of convergence. You can even receive your daily newspaper on your printer at home

electrical or electronic device via LCD touch panels and infrared or RF controllers. You can get a feel of how this works by visiting AMX's Web site ([www.amxcorp.com/solutions/index\\_solutions.asp](http://www.amxcorp.com/solutions/index_solutions.asp)). You can also program the system to create sequences of events, such as switching on the lights in a room when the doors to it are opened.

Remember those extra long days of work you put in at the office? And when you returned home, you had to face that mounting pile of unwashed clothes, the refrigerator running low on your favourite food groups and the house looking like a pigsty? Those days might be old lore, going by the improvements in processor manufacturing technology. Advancements in this field have made it possible to insert extremely small and powerful processors into almost any device. This means devices that use these processors can now perform tasks beyond their basic functionality. This basic level of 'intelligence' is now being incorporated into many household appliances. LG, for example, has an Internet refrigerator that will be able to alert users when expiry dates for items stored inside are approaching. It will also be able to order groceries over the Internet when supplies begin to run low.

Another example of this intelligence would be the Internet-ready washing machine, which can download wash programs for new types of fabrics from the Internet. In time, you will be able to remotely monitor these machines to program wash cycles. Even service technicians could monitor them remotely to diagnose problems. Down the road, these appliances will automatically detect each other and collaborate without any manual intervention.

The introduction of broadband networks that connect people's homes means that it is now fea-

sible for services to be delivered directly to you at home. Imagine the possibilities. Take the newspaper for instance. Most of us don't read it from one end to the other—we have our favourite sections.

Aside from the PC, which will continue to provide users with several ways to communicate, there are already some appliances such as InnoMedia's InfoTalk that allow you to make phone calls over the Internet using VoIP technology without requiring a PC. Videophones also can now operate off the same networks that bring Internet access into our homes. In time, we will probably be able to switch to an IP-based phone service, which will allow us to make voice, and videophone calls to anyone in the world.

Entertainment also seems to be moving in the same direction. Intelligent TV services like

TiVo, available in the US, automatically find and record your favourite programs and allow you to watch them at your convenience. While it currently uses a phone line to access TV programming details, eventually much greater functionality will be available to users of these kinds of services. An entertainment network in your home will make music, movies and the Internet accessible at any location in your house at all times. You can watch DVD movies, play games and surf the Internet with this single entertainment console.



Speak with IP-phone

### All under one roof

Audio-visual integrators and traditional electrical integrators are providing most of the existing home automation technology. They have formed consortiums like HAVi (Home Audio Video Interoperability, [www.havi.org](http://www.havi.org)) that provide a common, openly licensable specification for networking digital home entertainment products. However, there is

## Artificial Intelligence

It is the study of how computer systems can simulate intelligent processes such as learning, reasoning, and understanding symbolic information in context. Simply put, it is a branch of computer science concerned with making computers behave like humans. A new movie called AI, directed by Steven Spielberg, depicts a society that is largely dependent on intelligent robots and computer systems. While this might be far from becoming our reality, there are several areas where AI has made considerable progress.

AI involves work in algorithms, databases, and theoretical computer science as well as fields such as neuroscience, cognitive science and cognitive psychology, mathematical logic and engineering. Some areas wherein considerable progress has been made include natural language processing applications, robotics, and computer games. One of the most successful demonstrations of AI took place way back in May 1997, when a super-computer called Deep Blue from IBM defeated world champion Gary Kasparov in a chess match.